

4(a)	The principle of superposition states that when two or more travelling waves of the same type meet at a point in space, the resultant displacement at that point is the vector sum of the displacements due to each of the waves at that point.	
4(b)(i)	Resultant displacement = $1.8 - 2.6$ = $-0.8 \times 10^{-4} \text{ cm}$	
4(b)(ii)	$\frac{I}{I'} = \left(\frac{3}{1}\right)^2$ $I' = 0.11I$	
4(c)(i)	Path diff = $\sqrt{20^2 + 31.7^2} - \sqrt{20^2 + 28.7^2} = 2.5 \text{ m}$ Since P is third minima, $2.5 = \frac{5}{2} \lambda$ $\lambda = 1.0 \text{ m}$ $v = f \lambda$ $= \left(\frac{1}{3.0 \times 10^{-3}}\right)(1.0)$ $= 330 \text{ m/s}$	
4(c)(ii)	The intensity varies through 6 cycles of maximum and minimum.	
5(a)	The root mean square (r.m.s.) voltage of an alternating	